Unit - 4

Applied statistics

Type - I Fit a straight line

Let the best fit of the straight line be

$y = ax + b$.

The normal eqns are

$$a\sum x + nb = \sum y$$

$$a\sum x^2 + b\sum x = \sum xy$$

①. Fit a straight line to the data given below by the method of least square

x:	0	1	2	3	4
y:	1.0	1.8	3.3	4.5	6.3

Solns

Let the best fit of the line be $y = ax + b$ — Ⓐ

The normal eqns are

$$a\sum x + nb = \sum y$$

$$a\sum x^2 + b\sum x = \sum xy$$

x	y	x^2	xy
0	1	0	0
1	1.8	1	1.8
2	3.3	4	6.6
3	4.5	9	13.5
4	6.3	16	25.2
10	16.9	30	47.1

where $n=5$, $\Sigma x = 10$, $\Sigma y = 16.9$, $\Sigma x^2 = 30$

$\Sigma xy = 47.1$

Sub in normal eqns

$10a + 5b = 16.9$ — ①

$30a + 10b = 47.1$ — ②

Solving eqn ① & ② we get

$a = 1.33$ and $b = 0.72$

sub a & b values in Ⓐ

$y = ax + b$

$$\boxed{y = 1.33x + 0.72}$$

//

2. Find by the method of least squares the straight line that tests of fits the data on the following data.

x :	0	5	10	15	20
y :	7	11	16	20	26

calculate the value of y when $x = 23$.

Soln:

Let the best fit of the line be

$$y = ax + b \quad \text{—} \quad (A)$$

The normal eqns are

$$a \Sigma x + nb = \Sigma y$$

$$a \Sigma x^2 + b \Sigma x = \Sigma xy$$

x	y	x^2	xy
0	7	0	0
5	11	25	55
10	16	100	160
15	20	225	300
20	26	400	520
50	80	750	1035

where $n = 5$, $\Sigma x = 50$

$\Sigma y = 80$, $\Sigma x^2 = 750$

$\Sigma xy = 1035$

sub in normal eqns

$$50a + 5b = 80$$

$$750a + 50b = 1035$$

Solving we get

$$a = 0.94$$

$$b = 6.6$$

Sub a & b values in (A) we get.

$$\boxed{y = 0.94x + 6.6}$$

//

Example 4.1.2. Fit a straight line to the following data by the method of least squares:

x :	5	10	15	20	25
y :	16	19	23	26	30

Apr.'13

Solution . Let the best fit of the line be $y = ax + b$. ...(1)

Then we determine the parameters a and b by using method of least squares. The normal equations are

$$a\Sigma x + nb = \Sigma y$$
$$a\Sigma x^2 + b\Sigma x = \Sigma xy$$

x	y	x^2	xy
5	16	25	80
10	19	100	190
15	23	225	345
20	26	400	520
25	30	625	750
75	114	1375	1885

Where $n = 5$, $\Sigma x = 75$, $\Sigma y = 114$, $\Sigma x^2 = 1375$ and $\Sigma xy = 1885$

Substitute these values in the normal equations we get

$$75a + 5b = 114$$
$$1375a + 75b = 1885$$

Solving the above equations we get $a = 0.7$ and $b = 12.3$. Substitute a and b values in (1) we get $y = 0.7x + 12.3$

2. Fit a second degree polynomial (parabola)

Let the best fit of the parabolic curve

$\boxed{y = ax^2 + bx + c}$ The normal eqns are

$$a\sum x^2 + b\sum x + nc = \sum y$$

$$a\sum x^3 + b\sum x^2 + c\sum x = \sum xy$$

$$a\sum x^4 + b\sum x^3 + c\sum x^2 = \sum x^2 y$$

② The job rating efficiency of an employee seems to be related to the number of weeks of employment

Job efficiency :	55	50	20	55	75	80	90	30	75	70
weeks of employment :	2	4	1	3	5	9	12	2	7	5

Fit a second degree parabola using least squares.

Solution :-

Let the best fit $\boxed{y = ax^2 + bx + c}$

Assume $\boxed{X = \frac{x - 60}{5}}$

$\therefore \boxed{Y = aX^2 + bX + c}$ ①

Normal Equations are

$$\textcircled{1} \times \Sigma \Rightarrow \quad \Sigma Y = a \Sigma X^2 + b \Sigma X + nc$$
$$\textcircled{1} \times \Sigma X \Rightarrow \quad \Sigma XY = a \Sigma X^3 + b \Sigma X^2 + c \Sigma X$$
$$\textcircled{1} \times \Sigma X^2 \Rightarrow \quad \Sigma X^2 Y = a \Sigma X^4 + b \Sigma X^3 + c \Sigma X^2$$

x	$Y = y$	$X = \frac{x-60}{5}$	X^2	X^3	X^4	XY	X^2Y
55	2	-1	1	-1	1	-2	2
50	4	-2	4	-8	16	-8	16
20	1	-8	64	-512	4096	-8	64
55	3	-1	1	-1	1	-3	3
75	5	3	9	27	81	15	45
80	9	4	16	64	256	36	144
90	12	6	36	216	1296	72	432
30	2	-6	36	-216	1296	-12	72
75	7	3	9	27	81	21	63
70	5	2	4	8	16	10	20
	$\Sigma Y = 50$	$\Sigma X = 0$	$\Sigma X^2 = 180$	-396	7140	180	-396

Sub the values in Normal Equations

$$180a + 0b + 10c = 50$$
$$-396a + 180b + 0c = 121$$
$$7140a - 396b + 180c = 861$$

$$\boxed{a = 0.0750, \quad b = 0.8373, \quad c = 3.6498}$$

$$\therefore Y = 0.0750X^2 + 0.8373X + 3.6498$$

$$y = 0.0750\left(\frac{x-60}{5}\right)^2 + 0.8373\left(\frac{x-60}{5}\right) + 3.6498$$

$$y = \overset{0.003}{\cancel{0.0750}}\frac{(x-60)^2}{\cancel{25}} + \overset{0.1675}{\cancel{0.8373}}\frac{(x-60)}{\cancel{5}} + 3.6498$$

$$y = 0.003\left[x^2 - 120x + 3600\right] + 0.1675x - 10.05 + 3.6498$$

$$\boxed{y = 0.003x^2 - .1925x + 4.3998}$$

③ Fit a second degree parabola to the foll data and estimate value of the year 1990

year :	1955	1960	1965	1970	1975	1980	1985
Production :	6	8	9	10	12	11	8

Solution . Let x be year and y be the production.

Let the best be $y = ax^2 + bx + c$. ...(1)

Then we determine the parameters a and b by using method of least squares. The normal equations are

$$\begin{aligned} a\Sigma X^2 + b\Sigma X + nc &= \Sigma Y \\ a\Sigma X^3 + b\Sigma X^2 + c\Sigma X &= \Sigma XY \\ a\Sigma X^4 + b\Sigma X^3 + c\Sigma X^2 &= \Sigma X^2 Y \end{aligned}$$

x	$y = Y$	$X = \dfrac{x - 1970}{5}$	X^2	X^3	X^4	XY	X^2Y
1955	6	−3	9	−27	81	−18	54
1960	8	−2	4	−8	16	−16	32
1965	9	−1	1	−1	1	−9	9
1970	10	0	0	0	0	0	0
1975	12	1	1	1	1	12	12
1980	11	2	4	8	16	22	44
1985	8	3	9	27	81	24	72
	64	0	28	0	196	15	223

Where $n = 7$, $\Sigma X = 0$, $\Sigma Y = 64$, $\Sigma X^2 = 28$, $\Sigma X^3 = 0$, $\Sigma X^4 = 196$, $\Sigma XY = 15$ and $\Sigma X^2 Y = 223$

Substitute these values in the above normal equations we get

$$\begin{aligned} 28a + b.0 + 7c &= 64 \\ a.0 + 28b + c.0 &= 15 \\ 196a + b.0 + 28c &= 223 \end{aligned}$$

Solving the above equations we get $a = -0.3929$, $b = 0.5357$ and $c = 10.7143$ Substitute a, b and c values in (1) we get

$$\begin{aligned} Y &= aX^2 + bX + c \\ \therefore\ y &= -0.3929\left(\frac{x - 1970}{5}\right)^2 + 0.5357\left(\frac{x - 1970}{5}\right) + 10.7143 \end{aligned}$$

When $x = 1990$ then from the above parabolic fit we have

$$\begin{aligned} y &= -0.3929\,(4)^2 + 0.5357\,(4) + 10.7143 \\ y &= 6.5707 \end{aligned}$$

Example 4.1.8. Fit a second degree parabola of the form $y = ax^2 + bx + c$ to the following data.

x :	5	10	15	20	25	30	35
y :	15	21	35	40	37	23	12

Apr.'10

Solution . Let x be year and y be the production.

Let the best be $y = ax^2 + bx + c$. ...(1)

Then we determine the parameters a and b by using method of least squares. The normal equations are

$$\begin{aligned} a\Sigma X^2 + b\Sigma X + nc &= \Sigma Y \\ a\Sigma X^3 + b\Sigma X^2 + c\Sigma X &= \Sigma XY \\ a\Sigma X^4 + b\Sigma X^3 + c\Sigma X^2 &= \Sigma X^2 Y \end{aligned}$$

x	y	$Y = y - 25$	$X = \frac{x-20}{5}$	X^2	X^3	X^4	XY	X^2Y
5	15	−10	−3	9	−27	81	−150	−90
10	21	−4	−2	4	−8	16	−84	−16
15	35	10	−1	1	−1	1	350	10
20	40	15	0	0	0	0	600	0
25	37	12	1	1	1	1	444	12
30	23	−2	2	4	8	16	−46	−8
35	12	−13	3	9	27	81	−156	−117
		8	0	28	0	196	958	−209

Where $n = 7$, $\Sigma X = 0$, $\Sigma Y = 8$, $\Sigma X^2 = 28$, $\Sigma X^3 = 0$, $\Sigma X^4 = 196$, $\Sigma XY = 958$ and $\Sigma X^2 Y = -209$

Substitute these values in the above normal equations we get

$$
\begin{aligned}
28a + b.0 + 7c &= 8 \\
a.0 + 28b + c.0 &= 958 \\
196a + b.0 + 28c &= -209
\end{aligned}
$$

Solving the above equations we get $a = -2.869$, $b = 34.214$ and $c = 12.619$. Substitute a, b and c values in (1) we get

$$
\begin{aligned}
Y &= -2.869X^2 + 34.214X + 12.619 \\
\therefore\ (y - 25) &= -2.869\left(\frac{x-20}{5}\right)^2 + 34.214\left(\frac{x-20}{5}\right) + 12.619.
\end{aligned}
$$

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3. Fit a exponential curve

Let the n pairs of observations say (x_i, y_i) : $i = 1$ to n, then the best fit of the exponential curve $y = ae^{bx}$. ...(1)

Take logarithms (with base 10)on both sides we get

$$\begin{aligned}
\log_{10} y &= \log_{10}\left(ae^{bx}\right) \\
&= \log_{10} a + \log_{10} e^{bx} \\
\therefore \quad \log_{10} y &= \log_{10} a + xb\log_{10} e \\
\text{i.e., } \; Y &= A + Bx \qquad \text{...(2)}
\end{aligned}$$

In (2), we determine the parameters A and B by using the given normal equations,

$$\begin{aligned}
nA + B\Sigma x &= \Sigma Y \\
A\Sigma x + B\Sigma x^2 &= \Sigma xY
\end{aligned}$$

Solving the above two equations, we get the values for A and B.

$$\begin{aligned}
\text{Since, } \; \log_{10} a &= A. \;\Rightarrow a = \text{antilog}\, A \;\; \text{(or)} \;\; 10^A. \\
\text{and } \; b\log_{10} e &= B. \;\Rightarrow b = \frac{B}{\log_{10} e} \\
&= \frac{B}{0.4343} \qquad [\because \log_{10} e = 0.4343]
\end{aligned}$$

Substitute a and b values in (1), we get the required exponential fit.

① Fit a curve of the form $y = ae^{bx}$ for the following data:

x : 1 5 7 9 12

y : 10 15 12 15 21

<u>soln:</u> Let the best fit of the line be

$$y = ae^{bx} \quad — ①$$

Taking log on both sides

$$\log_{10} y = \log_{10}(ae^{bx})$$

$$= \log_{10} a + \log_{10} e^{bx}$$

$$\therefore \log_{10} y = \log_{10} a + bx \log_{10} e \quad — ②$$

$$Y = A + Bx$$

Then we determine the parameters A and B by using method of least squares.

The normal eqns are

$$nA + B\Sigma x = \Sigma Y$$

$$A\Sigma x + B\Sigma x^2 = \Sigma xY$$

x	y	$Y = \log_{10} y$	x^2	xY
1	10	1.000	1	1
5	15	1.1761	25	5.8805
7	12	1.0792	49	7.5544
9	15	1.1761	81	10.5849
12	21	1.3222	144	15.8664
34		5.7536	300	40.8862

Substitute these values in the normal eqns we get

$$5A + 34B = 5.7536$$

$$34A + 300B = 40.8862$$

Solving above eqns we get

$$A = 0.9766 \quad \& \quad B = 0.02561$$

$\therefore a = \text{antilog } A = 9.4754$ | $b = \dfrac{B}{\log_{10} e} = 0.059$

$$\therefore y = ae^{bx}$$

$$\therefore y = 9.4754\, e^{0.059x}$$

4. Fit a curve of the form $y = ab^x$

Let the n pairs of observations say (x_i, y_i) : $i = 1$ to n, then the best fit of the curve $y = ab^x$. ...(1)

Take logarithms (with base 10)on both sides we get

$$\begin{aligned}
\log_{10} y &= \log_{10}(ab^x) \\
&= \log_{10} a + \log_{10} b^x \\
\therefore \log_{10} y &= \log_{10} a + x \log_{10} b \\
\text{i.e.,}\ Y &= A + Bx \qquad ...(2)
\end{aligned}$$

In (2), we determine the parameters A and B by using the given normal equations,

$$\begin{aligned}
nA + B\Sigma x &= \Sigma Y \\
A\Sigma x + B\Sigma x^2 &= \Sigma xY
\end{aligned}$$

Solving the above two equations, we get the values for A and B.

$$\begin{aligned}
\text{Since,}\quad \log_{10} a &= A. \Rightarrow a = \text{antilog}\, A \ \text{(or)}\ 10^A. \\
\text{and}\quad \log_{10} b &= B. \Rightarrow b = \text{antilog}\, B \ \text{(or)}\ 10^B.
\end{aligned}$$

Substitute a and b values in (1), we get the required fit $y = ab^x$.

5 Fit a curve of the form $y = ax^b$

Let the n pairs of observations say (x_i, y_i) : $i = 1$ to n, then the best fit of the curve $y = ax^b$. ...(1)

Take logarithms (with base 10)on both sides we get

$$\begin{aligned}
\log_{10} y &= \log_{10}\left(ax^b\right) \\
&= \log_{10} a + \log_{10} x^b \\
\therefore \log_{10} y &= \log_{10} a + b \log_{10} x \qquad ...(2) \\
\text{i.e.,}\ Y &= A + bX
\end{aligned}$$

In ②, we determine the parameters A and b by using the given normal eqns,

$$nA + b\Sigma X = \Sigma Y$$

$$A\Sigma X + b\Sigma X^2 = \Sigma XY$$

Solving two eqns, we get the values for A and b. Find the value of a from A then substituting a and b values in the eqn $y = ax^b$ we get the required fit of the curve.

① . obtain a relation of the form $y = ab^x$ for the following data by the method of least squares.

x:	2	3	4	5	6
y:	8.3	15.4	33.1	65.2	127.4

Soln:

Let the best fit of the line be $y = ab^x$ — ①

Taking log on both sides of ①, we get

$$\log_{10} y = \log_{10}(ab^x)$$

$$= \log_{10} a + \log_{10} b^x$$

$$\therefore \log_{10} y = \log_{10} a + x \log_{10} b \quad \text{—— ②}$$

$$Y = A + Bx$$

The normal eqns are

$$nA + B\Sigma x = \Sigma Y$$

$$A\Sigma x + B\Sigma x^2 = \Sigma xY$$

x	y	$Y = \log_{10} y$	x^2	xY
2	8.3	0.9191	4	1.8382
3	15.4	1.1872	9	3.5616
4	33.1	1.5198	16	6.0792
5	65.2	1.8142	25	9.0710
6	127.4	2.1052	36	12.6312
20		7.545	90	33.1812

where $n=5$, $\Sigma x = 20$, $\Sigma Y = 7.545$,

$\Sigma x^2 = 90$ and $\Sigma xY = 33.1812$

Sub in normal eqns

$$5A + 20B = 7.545$$
$$20A + 90B = 33.18112$$

Solve above eqns we get $A = 0.31$ and $B = 0.3$

$\therefore a = \text{antilog } A$ and $b = \text{antilog } B = 1.995$

$\boxed{a = 2.04}$ $\boxed{b = 1.995}$

Thus the required soln is

$$y = ab^x$$

$$\therefore y = 2.04\,(1.995)^x.$$

2. Fit a curve of the form $y = ax^b$ to the given data:

x:	1	2	3	4	5
y:	7.1	27.8	62.1	110	161

Soln:

Let the best fit of the line be $y = ax^b$ —— (1)

Taking log on both sides of (1), we get

$$\log_{10} y = \log_{10} (ax^b)$$

$$= \log_{10} a + \log_{10} x^b$$

$$\therefore \log_{10} y = \log_{10} a + b \log_{10} x \quad \text{—— (2)}$$

$$Y = A + bX$$

The normal eqns are

$$nA + b\sum X = \sum Y$$

$$A\sum X + b\sum X^2 = \sum XY$$

x	y	$Y = \log_{10} y$	$X = \log_{10} x$	X^2	XY
1	7.1	0	0.8513	0	0
2	27.8	0.301	1.444	0.0906	0.4347
3	62.1	0.4771	1.7931	0.2276	0.8555
4	110	0.6021	2.0414	0.3625	1.229
5	161	0.699	2.2068	0.4886	1.5425
15		20.0792	8.3366	1.1693	4.0618

where $n=5$, $\quad \Sigma x = 2.0792$

$\Sigma y = 8.3366$, $\quad \Sigma x^2 = 1.1693$

$\Sigma xy = 4.0618$

Sub in normal eqns we get

$5A + 2.0791b = 8.3366$

$2.07916A + 1.1692b = 4.0618$

Solving above eqns we get

$A = 0.85494$

and $\quad b = 1.9536$.

$\therefore a = \text{antilog } A$

$\boxed{a = 7.16044}$

$\boxed{b = 1.9536}$

$\therefore$ The required soln is

$y = ax^b$

$y = 7.16044\, x^{(1.9536\,x)}$

4.3 Test of Significance of Large Samples

If the sample size $n \geq 30$, then the samples are large samples. In the large samples, the distributions, such as Binomial, Poisson, Chi-square etc. are closely approximated to normal distribution. Therefore, we apply normal test for large samples then by assuming the population as normal.

Under the test of large samples, we shall consider some following important tests to test the significance.

1. Test of significance for single proportion.
2. Test of significance for difference of proportions.
3. Test of significance for single mean.
4. Test of significance for difference of means.
5. Test of significance for a standard deviation.
6. Test of significance for difference of standard deviations.

Test of Significance of Single Proportion

$p = \frac{x}{n}$ $\boxed{x \to \text{no of items in a drawn sample } n}$

$P \to$ Population proportion

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}$$ where $Q = 1 - P$

$\because \boxed{P + Q = 1}$

Level of Significance

$\boxed{\text{one tail}}$ 1% $\Rightarrow$ 2.33, $\boxed{5\% = 1.64}$ (✓), 10% = 1.28

$\boxed{\text{two tail}}$ 1% $\to$ 2.58, $\boxed{5\% = 1.96}$ $\boxed{10\% = 1.64}$

① In a big city 325 men out of 600 men were found to be smokers. Does the information support the conclusion that the majority of men in this city are smokers?

Solution

$n = 600$, $x = 325$

$$p = \frac{x}{n} = \frac{325}{600} = 0.542$$

P = population proportion of smokers $= \frac{1}{2}$

$\therefore$ $Q = 1 - P = 1 - \frac{1}{2} = \frac{1}{2}$

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Null Hypothesis : (H_0) = Majority of men in the city are smokers.

Calculated Value

$$z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.542 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{600}}} = \frac{0.042}{\sqrt{\frac{1}{2400}}}$$

$$\boxed{z = 2.04}$$

$\therefore$ calculated value $z = 2.04$ (CV)

Tabulated value (TV)

Level of significance 5% = 1.96

$\because$ CV > TV

Reject H_0

(2)

1) In a sample of 1000 peoples in Maharashtra 540 are rice eaters and rest are wheat eaters. Can assume that both rice and wheat are equally popular. In this state and 1% level of significance.

$n = 1000$

$p = \frac{x}{n} = \frac{540}{1000} = 0.54$

P = population proportion of rice eaters in maharashtra

$P = \frac{1}{2}$

$Q = 1 - P = 1 - \frac{1}{2} = 1 - 0.5 = 0.5$

The parameter of interest is p

Null hypothesis $H_0 : p = P(0.5)$

Alternative hypothesis $H_1 \to p \neq P(0.5)$

Level of significance $\alpha = 1\% = \frac{1}{100} = 0.01$

Test of statistics $z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}$

$= \frac{0.54 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{1000}}} = 2.529$

Conclusion :-

$\because |z| = 2.529 < 2.58$ so we accept H_0 at 1% level of significance

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$$= (0.05413, 0.1059).$$

Example 4.3.13. In a city, a sample of 1000 people was taken and out of them 540 are vegetarians and the rest are non-vegetarians. Can we say that both habits of eating [vegetarian (or) non-vegetarian] are equally popular in the city at (i) 1% level of significance. (ii) 5% level of significance.

Apr.'14

Solution . Step 1: Sample $n = 1000$. Number of vegetarians $= 540$.

p = Sample proportion of vegetarians $= \frac{540}{1000} = 0.54$

P = Population proportion of vegetarians $= \frac{1}{2} = 0.5$

$\therefore Q = 1 - P = 0.5$

Step 2 : Null hypothesis (H_0) : $P = \frac{1}{2}$. i.e., Both vegetarians and non-vegetarians are equal in the city.

Alternative hypothesis (H_1) : $P \neq \frac{1}{2}$. (Two tailed).

Step 3: Calculated value of z :

$$\text{Test statistic } z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.54 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{1000}}} = 2.529$$

$\therefore$ The calculated value of $z = 2.529$

dec,
2, 7, more

same eq

Two tailed One tailed

Step 4: Tabulated value of z at

(i) 1% level of significance = 2.58 =2.33

(ii) 5% level of significance = 1.96 =1.645

Step 5: Conclusion :

(i) At 1% level, the calculated value of $z <$ the tabulated value of z. $\therefore$ Accept H_0. The means that there is a significance difference between vegetarians and non-vegetarians.

(ii) At 5% level, the calculated value of $z >$ the tabulated value of z. $\therefore$ Reject H_0. The means that no difference between vegetarians and non-vegetarians.

Example 4.3.14. In a city, a sample of 500 people, 280 are tea drinkers and the rest are coffee drinkers. Can we assume that ... this city at 5% level of

Example 4.3.9. 10 dice are thrown 500 times. Getting 2, 3 or 4 is considered a successes obtained in 2560 times. Can this be attributed to fluctuations in sampling.

Solution . Step 1: $n = 500$. This experiment is repeated for 10 times. $\therefore$ Total samples are $10 \times 500 = 5000$.

Number of Success = 2560.

p = sample proportion of success $= \frac{2560}{5000} = 0.512$

P = Population proportion of success $= P(\text{get } 2, 3 \text{ (or) } 4)$

$= \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2} = 0.5. \quad \therefore Q = 1 - P = 0.5$

Step 2 : Null hypothesis (H_0) : The die is unbiased. or $P = \frac{1}{2}$

Alternative hypothesis (H_1) : $P \neq \frac{1}{2}$. (i.e., Two tailed).

Step 3: Calculated value of z :

$$\text{Test statistic } z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.512 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{500}}} = 1.697$$

$\therefore$ The calculated value of $z = 1.697$

Step 4: Tabulated value of z at 5% level of significance = 1.96

Step 5: Conclusion :

Since calculated value of z < tabulated value of z.

$\therefore$ Accept H_0. The means that the difference could be attributed due to fluctuations of sampling.

Example 4.3.10. A coin is tossed 400 times and it turns up head 216 times. Discuss whether the coin may be regarded as unbiased one.

Nov.'12

Solution . Step 1: Sample size $n = 400$. Number of heads $= 216$.

$p =$ sample proportion of heads $= \frac{216}{400} = 0.54$

$P =$ Population proportion of success $= 0.5 \therefore Q = 1 - P = 0.5$

Step 2 : Null hypothesis (H_0) : Coin is unbiased. (or) $P = \frac{1}{2}$

Alternative hypothesis (H_1) : $P \neq \frac{1}{2}$..

Step 3: Calculated value of z :

$$\text{Test statistic } z = \frac{p-P}{\sqrt{\frac{PQ}{n}}} = \frac{0.54-0.5}{\sqrt{\frac{0.5\times0.5}{400}}} = 1.6$$

$\therefore$ The calculated value of $z = 1.6$

Step 4: Tabulated value of z at 5% level of significance $= 1.96$

Step 5: Conclusion :

Since calculated value of $z <$ tabulated value of z.

$\therefore$ Accept H_0. Thus coin is unbiased.

Example 4.3.11. A die is thrown 9000 times and a throw of 3 or 4 is reckoned as a success. Suppose that 3240 throws of a 3 or 4 have been made out. Do the data indicate an unbiased die? If not, find the probable limits of probability of getting 3 or 4.

Solution . Step 1: Sample size $= 9000$. Number of Success $= 3240$.

$p =$ sample proportion of success $= \frac{3240}{9000} = 0.36$

$P =$ Population proportion of success $= P(\text{get 3 or 4})$

$= \frac{1}{6} + \frac{1}{6} = \frac{1}{3} = 0.33. \therefore Q = 1 - P = 0.67$

Step 2 : Null hypothesis (H_0) : The die is unbiased. (or) $P = \frac{1}{2}$

Alternative hypothesis (H_1) : $P \neq \frac{1}{2}$.

Step 3: Calculated value of z :

$$\text{Test statistic } z = \frac{p-P}{\sqrt{\frac{PQ}{n}}} = \frac{0.36-0.5}{\sqrt{\frac{0.5\times0.5}{9000}}} = 6.04$$

$\therefore$ The calculated value of $z = 6.04$

Step 4: Tabulated value of z at 5% level of significance $= 1.96$

Step 5: **Conclusion :**

Since calculated value of $z >$ tabulated value of z.

$\therefore$ Reject H_0. The means that the die is biased.

To find limits of population proportion

Probable limits of probability of getting 3 or 4

$$= P \pm 1.96\sqrt{\frac{PQ}{n}}. \quad \text{Where } Q = 1 - P.$$

$$= 0.36 \pm 1.96\sqrt{\frac{0.33 \times 0.67}{9000}} = 0.36 \pm 1.96(0.005)$$

$$= (0.3502,\ 0.3698)$$

4.3.2 Test of Significance of Difference of Two Proportions

Let n_1 and n_2 be the number of two drawn samples from the population. Let x_1 and x_2 be the number of items/individuals possessing a certain kind of attribute in the drawn samples n_1 and n_2. Then the sample proportions are calculated by $p_1 = \frac{x_1}{n_1}$ and $p_2 = \frac{x_2}{n_2}$. If the population proportion P is not known, then the significance between the sample proportions p_1 and p_2 by using test statistic is given by

$$z = \frac{p_1 - p_2}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}.$$

$$\text{Where } P = \frac{n_1p_1 + n_2p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} \text{ and } Q = 1 - P.$$

Note :

Standard error of difference of proportions $S.E. = \sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$

Working rule :

1. Write samples n_1 and n_2 and sample proportions p_1 and p_2.
2. State null hypothesis H_0 and alternative H_1.
3. Find the calculated value of z by using test statistic

$$z = \frac{p_1 - p_2}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \text{ where } P = \frac{n_1p_1 + n_2p_2}{n_1 + n_2}, \quad Q = 1 - P$$

and consider $|z|$ if z becomes negative.

4. Write tabulated value of z at 1% or 5% level of significance.
5. Conclusion :

(i) If the calculated value of $z <$ tabulated value of z, then accept H_0 (or)

(ii) If the calculated value of $z >$ tabulated value of z, then reject H_0.

Example 4.3.25. In a random sample of 1000 persons from the city Coimbatore. 400 are found to be consumers of wheat. In a sample of 800 from the city Madurai, 400 are found to be consumers of wheat. Do these data reveal a significant difference between the two cities, so far as the proportion of wheat consumers in concerned? **Apr.'14**

Solution . Step 1: Given samples size $n_1 = 1000,\ n_2 = 800.$

$$p_1 = \frac{400}{1000} = 0.4 \text{ and } p_2 = \frac{400}{800} = 0.5$$

$$\therefore\ P = \frac{n_1p_1 + n_2p_2}{n_1 + n_2} = \frac{1000\,(.4) + 800\,(.5)}{1000 + 800}$$
$$= 0.44 \quad \therefore\ Q = 1 - P = 0.56$$

Step 2 : Null hypothesis (H_0) : There is no significance difference in the consumers of wheat in the two cities. i.e., $P_1 = P_2$.

Alternative hypothesis (H_1) : $P_1 \neq P_2$.

Step 3: Calculated value of z :

$$\text{Test statistic } z = \frac{p_1 - p_2}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$
$$= \frac{0.4 - 0.5}{\sqrt{0.44 \times 0.56\left(\frac{1}{1000} + \frac{1}{800}\right)}}$$
$$= -4.247$$

$\therefore$ The calculated value of $|z| = 4.247$

Step 4: Tabulated value of z at 5% level of significance $= 1.96$

Step 5: Conclusion :

At 5% level, the calculated value of $z >$ tabulated value of z.
$\therefore$ Reject H_0. There is a significant of the proportion of wheat consumers in the two cities.

Example 4.3.24. In a large city A, 20% of a random sample of 900 school boys had defective eye-sight. In another large city B, 18.5% of a random sample of 1600 school boys had the same defect. Is the difference between the two proportions significant? **Nov.'06**

Solution . Step 1: Given samples size $n_1 = 900$, $n_2 = 1600$.

$$p_1 = 20\% = 0.2 \text{ and } p_2 = 18.5\% = 0.185$$

$$\therefore P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{900\,(.2) + 1600\,(.185)}{900 + 1600}$$

$$= 0.19 \qquad \therefore Q = 1 - P = 0.81$$

Step 2 : Null hypothesis (H_0) : There is no significance difference between the proportions. i.e.,$P_1 = P_2$.

Alternative hypothesis (H_1) : $P_1 \neq P_2$.

Step 3: Calculated value of z :

$$\text{Test statistic } z = \frac{p_1 - p_2}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$= \frac{0.2 - 0.185}{\sqrt{0.19 \times 0.81\left(\frac{1}{900} + \frac{1}{1600}\right)}}$$

$$= 0.918$$

Step 4: Tabulated value of z at 5% level of significance $= 1.96$

Step 5: Conclusion :

At 5% level, the calculated value of $z <$ tabulated value of z.

$\therefore$ Accept H_0. The difference between the proportions are not significant.

Example 4.3.17. Before an increase in excise duty on tea, 800 persons, out of a sample of 1000 persons were found to be tea drinkers. After an increase in excise duty, 800 were tea drinkers in a sample of 1200 people. State whether there is a significant decrease in the consumption of tea after the increase in excise in excise duty?

Nov.'11,Apr.'15

Solution . Step 1: Given samples size $n_1 = 1000, \ n_2 = 1200.$

$$p_1 = \frac{800}{1000} = .8 \text{ and } p_2 = \frac{800}{1200} = 0.667$$

$$\therefore P = \frac{n_1p_1 + n_2p_2}{n_1 + n_2} = \frac{1000\,(.8) + 1200\,(.667)}{1000 + 1200}$$

$$= 0.727 \qquad \therefore Q = 1 - P = 0.273$$

Step 2 : Null hypothesis (H_0) **:** Proportion of tea drinkers before and after the increase in excise duty are equal. i.e.,$P_1 = P_2$.

Alternative hypothesis (H_1) **:** $\cancel{P_1 >} P_2$. $P_1 \neq P_2$

Step 3: Calculated value of z :

$$\text{Test statistic } z = \frac{p_1 - p_2}{\sqrt{PQ\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}}$$
$$= \frac{0.8 - 0.667}{\sqrt{0.727 \times 0.273\left(\frac{1}{1000}+\frac{1}{1200}\right)}}$$
$$= \frac{0.133}{0.019078}$$
$$= 6.971$$

$\therefore$ The calculated value of $z = 6.971$

1.96

Step 4: Tabulated value of z at 5% level of significance $= \cancel{1.645}$

Step 5: Conclusion :

At 5% level, the calculated value of $z >$ tabulated value of z.
$\therefore$ Reject H_0. There is a significant decrease in the consumption of tea due to increase in excise duty.

Example 4.3.18. In a sample of 300 units of a manufactured product, 65 units were found to be defective and in another sample of 200 units, there were 35 defectives. Is there significant difference in the proportion of defectives in the samples at 5% level of significance.

Solution . Step 1: Given samples size $n_1 = 300,\ n_2 = 200$.

$$p_1 = \frac{65}{300} = 0.22 \text{ and } p_2 = \frac{35}{200} = 0.175$$
$$\therefore P = \frac{n_1p_1 + n_2p_2}{n_1+n_2} = \frac{300\,(.22) + 200\,(.175)}{300+200}$$
$$= 0.2 \quad \therefore Q = 1 - P = 0.8$$

Step 2 : Null hypothesis (H_0) : There is no significance difference in the proportion of defectives in the samples. i.e., $P_1 = P_2$.
Alternative hypothesis (H_1) : $P_1 \neq P_2$.

Step 3: Calculated value of z :

$$\text{Test statistic } z = \frac{p_1 - p_2}{\sqrt{PQ\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}} = \frac{0.22 - 0.175}{\sqrt{0.2 \times 0.8\left(\frac{1}{300}+\frac{1}{200}\right)}}$$

$$= 1.233$$

Step 4: Tabulated value of z at 5% level of significance $= 1.96$

Step 5: Conclusion :

At 5% level, the calculated value of z < tabulated value of z.

$\therefore$ Accept H_0. The difference in the proportion of defectives in the samples is not significant.

Example 4.3.19. A machine puts out 16 imperfect articles in a sample of 500. After the machine is overhauled, it puts out 3 imperfect articles in a batch of 100. Has the machine improved. **Apr.'11**

Solution . Step 1: Given samples size $n_1 = 500,\ n_2 = 100.$

$$p_1 = \frac{16}{500} = 0.032 \text{ and } p_2 = \frac{3}{100} = 0.03$$

$$\therefore P = \frac{n_1p_1 + n_2p_2}{n_1+n_2} = \frac{500\,(.032) + 100\,(.03)}{500+100}$$

$$= 0.0317 \quad \therefore Q = 1 - P = 0.9683$$

Step 2 : Null hypothesis (H_0) : Machine is not improved before and after over haulding. i.e.,$P_1 = P_2$.

Alternative hypothesis (H_1) : ~~$P_1 > P_2$~~. $P_1 \neq P_2$

Step 3: Calculated value of z :

$$\text{Test statistic } z = \frac{p_1 - p_2}{\sqrt{PQ\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}}$$

$$= \frac{0.032 - 0.03}{\sqrt{0.0317 \times 0.9683\left(\frac{1}{500}+\frac{1}{100}\right)}}$$

CV $= 0.1042$ TV $= 1.96$

CV < TV

Accept H_0

4.4 Population and Single Mean

Let $x_1, x_2, ...x_n$ be the size of n sample observations/items drawn from the population with mean μ and S.D. σ. Let $\overline{x}$ be the sample mean. Assuming that the population is normal $N(\mu, \sigma)$. Then the test statistic for single mean with the population mean is given by

$$\boxed{z = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}}.}$$

If population S.D.(σ) is not known, then test statistic for the mean of single sample is given by

$$z = \frac{\overline{x} - \mu}{s/\sqrt{n}}.$$

Where s is S.D. of drawn sample.

Note :

1. Standard error of sample mean $\overline{x} = \frac{\sigma}{\sqrt{n}}$
2. 95% limits of population mean μ are $\overline{x} \pm 1.96\frac{\sigma}{\sqrt{n}}$, if σ is known.
3. 99% confidence limits for μ are $\overline{x} \pm 2.58\frac{\sigma}{\sqrt{n}}$.

Working rule :

1. Write the sample size (n), sample mean ($\overline{x}$), population mean (μ), population S.D. (σ).
2. State null hypothesis H_0 and alternative hypothesis H_1.
3. Find the calculated value of z by using test statistic $z = \frac{\overline{x}-\mu}{\sigma/\sqrt{n}}$. and consider $|z|$ if z becomes negative.
4. Write tabulated value of z at 1% or 5% level of significance.
5. Conclusion :

   (i) If the calculated value of z < tabulated value of z, then accept H_0 (or)

   (ii) If the calculated value of z > tabulated value of z, then reject H_0.

Example 4.4.1. The average marks in Mathematics of a sample of 100 students was 51 with a S.D. of 6 marks. Could this have been a random sample from a population with average marks 50? **Nov.'12**

Solution . Step 1: Given $n = 100$, $\overline{x} = 51$, $\mu = 50$ and $s = 6$.

Step 2 : Null hypothesis (H_0) : $\mu = 50$.

Alternative hypothesis (H_1) : $\mu \neq 50$.

Step 3: Calculated value of z :

$$\text{Test statistic } z = \frac{\overline{x}-\mu}{s/\sqrt{n}} = \frac{51-50}{6/\sqrt{100}}$$

$$\therefore z = 1.667$$

Step 4: Tabulated value of z at 5% level of significance $= 1.96$

Step 5: Conclusion :

At 5% level, the calculated value of $z <$ tabulated value of z.

$\therefore$ Accept H_0. There is no significance of the population mean and sample mean.

Example 4.4.2. A sample of 900 members has a mean of 3.4 cms and S.D. is 2.61 cms. Is the sample from a large population of mean 3.25 cm and S.D. is 2.61 cms. If the population is normal and its mean is unknown find the 95% confidence limits of true mean.

Apr.'06, Apr.'12

Solution . Step 1: Given $n = 900$, $\overline{x} = 3.4$, $\mu = 3.25$ and $\sigma = 2.61$

Step 2 : Null hypothesis (H_0) : $\mu = 3.25$.

Alternative hypothesis (H_1) : $\mu \neq 3.25$.

Step 3: Calculated value of z :

$$\text{Test statistic } z = \frac{\overline{x}-\mu}{\sigma/\sqrt{n}} = \frac{3.4-3.25}{2.61/\sqrt{900}}$$

$$\therefore z = 1.724$$

Step 4: Tabulated value of z at 5% level of significance $= 1.96$

Step 5: Conclusion :

At 5% level, the calculated value of $z <$ tabulated value of z.

$\therefore$ Accept H_0. There is no significance difference between the population mean and sample mean.

95% confidence limits of mean μ are

$$\begin{aligned} &= \overline{x} \pm 1.96 \frac{\sigma}{\sqrt{n}} \\ &= 3.4 \pm 1.96 \times \frac{2.61}{\sqrt{900}} \\ &= 3.4 \pm 0.17 \\ &= 3.23,\ 3.57 \end{aligned}$$

Example 4.4.3. 400 workers were selected at random from a district. Their mean income was Rs. 140.50 per month with standard deviation of Rs. 25.20. Do you believe that the average income of the workers community in the district is Rs. 150?

Solution . Step 1: Given $n = 400$, $\overline{x} = 140.50$, $\mu = 150$ and $\sigma = 25.2$

Step 2 : Null hypothesis (H_0) : $\mu = 150$.

Alternative hypothesis (H_1) : $\mu \neq 150$.

Step 3: Calculated value of z :

$$\text{Test statistic } z = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{140.50 - 150}{25.2/\sqrt{400}}$$

$$\therefore z = -7.54$$

$\therefore$ The calculated value of $|z| = 7.54$

Step 4: Tabulated value of z at 5% level of significance $= 1.96$

tion mean and sample mean.

Example 4.4.10. A random sample of 100 steel rods was drawn from a population of rods whose lengths are normally distributed with mean 4 feet and standard deviation 0.6 feet. If the sample mean is 4.2 feet. Can the sample be regarded as a random sample drawn from the above population?

Solution . Step 1: Given $n = 100$, $\overline{x} = 4.2$, $\mu = 4$ and $\sigma = 0.6$

Step 2 : Null hypothesis (H_0) : $\mu = 4$.

Alternative hypothesis (H_1) : $\mu \neq 4$.

Step 3: Calculated value of z :

$$\text{Test statistic } z = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{4.2 - 4}{0.6/\sqrt{100}} = 3.33$$

Step 4: Tabulated value of z at 5% level of significance $= 1.96$

Step 5: Conclusion :

At 5% level, the calculated value of $z >$ tabulated value of z.

$\therefore$ Reject H_0. Thus the sample is not a random sample drawn from the given population.

Example 4.4.11. A sample of 100 people during the past year showed an average life span of 71.8 years. If the standard deviation of the population is 8.9 years. Test whether the mean life span today is greater than 70 years. **Nov.'14, Nov.'15**

Solution . Step 1: Given $n = 100$, $\overline{x} = 71.8$, $\mu = 70$ and $\sigma = 8.9$.

Step 2 : Null hypothesis (H_0) : $\mu = 70$.

Alternative hypothesis (H_1) : $\mu > 70$.

Step 3: Calculated value of z **:**

$$\text{Test statistic } z = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{71.8 - 70}{8.9/\sqrt{100}} = 2.0224$$

Step 4: Tabulated value of z at 5% level of significance = ~~1.645~~ 1.96

Step 5: Conclusion :

At 5% level, the calculated value of z $\cancel{>}$ tabulated value of z.

$\therefore$ Reject H_0. Difference is significant.

Example 4.4.12. The heights of college students in a city are normally distributed with SD 6 cms. A sample of 100 students has mean height 158 cms. Test the hypothesis that the mean height of college students in the city is 160 cms. Also obtain 99% confidence limits for the true mean.

Apr.'14

Solution . Step 1: Given $n = 100$, $\overline{x} = 158$, $\mu = 160$ and $\sigma = 6$.

Step 2 : Null hypothesis (H_0) : $\mu = 160$.

Alternative hypothesis (H_1) : $\mu \neq 160$.

Step 3: Calculated value of z **:**

$$\text{Test statistic } z = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{158 - 160}{6/\sqrt{100}} = -3.33$$

$\therefore$ Calculated value of $|z| = 3.33$

Step 4: Tabulated value of z at 5% level of significance = 1.96

Step 5: Conclusion :

At 5% level, the calculated value of z > tabulated value of z.

$\therefore$ Reject H_0. Difference is significant.

99% confidence interval for μ are

$$= \bar{x} \pm 2.58 \frac{\sigma}{\sqrt{n}}$$

$$= 158 \pm 2.58 \frac{6}{\sqrt{100}}$$

$$= 158 \pm 1.548$$

$$= (156.452,\ 159.548)$$

$\therefore$ $156.452 < \mu < 159.548$. Here 160 does not lie in the interval.

mean 0.1 and S.D. 2.1.

4.4.1 Test of Significance of Difference of Two Means

Let n_1 and n_2 be the two samples drawn from two different populations with population means μ_1 and μ_2 and standard deviations σ_1 and σ_2. Let $\overline{x}_1$ and $\overline{x}_2$ be the two sample means follow normal distribution. Then the test statistic is

$$z = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

. **Note :** 1. If the samples are drawn from the same population with S.D. $\sigma_1 = \sigma_2 = \sigma$ then the test statistic is

$$z = \frac{\overline{x}_1 - \overline{x}_2}{\sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}.$$

2. If the populations standard deviations σ_1 and σ_2 are not given. In such case, the standard deviations of samples (i.e., s_1, s_2) are given then we substitute standard deviations of samples s_1, s_2 in the place of σ_1 and σ_2 then the test statistic is

$$z = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Note :

Standard error of difference of two means $(\overline{x}_1 - \overline{x}_2) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$.

Working rule :

1. Write the samples size n_1 and n_2, means $\overline{x}_1$ and $\overline{x}_2$, S.D. σ_1 and σ_2.
2. State null hypothesis H_0 and alternative hypothesis H_1.
3. Find the calculated value of z by using test statistic $z = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$. and consider $|z|$ if z becomes negative.
4. Write tabulated value of z at 1% or 5% level of significance.
5. Conclusion :

   (i) if the calculated value of $z <$ tabulated value of z, then accept H_0 (or)

   (ii) if the calculated value of $z >$ tabulated value of z, then reject H_0.

Example 4.4.15. The means of two simple samples of 1000 and 2000 items are 170 cm and 169 cm respectively. Can the samples be regarded as drawn from the same population with standard deviation 10 at 5% level of significance? **Apr.'12**

Solution . Step 1: Given $n_1 = 1000$, $n_2 = 2000$ and $\overline{x}_1 = 170$, $\overline{x}_2 = 169$ and $\sigma = 10$.

Step 2 : Null hypothesis (H_0) : Samples are drawn from the same population. i.e., $\mu_1 = \mu_2$

Alternative hypothesis (H_1) : $\mu_1 \neq \mu_2$.

Step 3: Calculated value of z :

$$\text{Test statistic } z = \frac{\overline{x}_1 - \overline{x}_2}{\sigma\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} = \frac{170-169}{10\sqrt{\frac{1}{1000}+\frac{1}{2000}}}$$

$$\therefore z = 2.589$$

Step 4: Tabulated value of z at 5% level of significance = 1.96

Step 5: Conclusion :

At 5% level, the calculated value of z > tabulated value of z.

$\therefore$ Reject H_0. The samples cannot be regarded from the same population.

Example 4.4.16. The mean height of two samples of 1000 and 2000 members are respectively 67.5 and 68 inches. Can they be regarded as drawn from the same population with standard deviation 2.5 inches.

Nov.'14

Solution . Step 1: Given $n_1 = 1000$, $n_2 = 2000$ and $\overline{x}_1 = 67.5$, $\overline{x}_2 = 68$ and $\sigma = 2.5$.

Step 2 : Null hypothesis (H_0) : Samples are drawn from the same population. i.e., $\mu_1 = \mu_2$

Alternative hypothesis (H_1) : $\mu_1 \neq \mu_2$.

Step 3: Calculated value of z :

$$\text{Test statistic } z = \frac{\overline{x}_1 - \overline{x}_2}{\sigma\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} = \frac{67.5-68}{2.5\sqrt{\frac{1}{1000}+\frac{1}{2000}}}$$

$$\therefore z = -5.164$$

$\therefore$ The calculated value of $|z| = 5.164$

Step 4: Tabulated value of z at 5% level of significance $= 1.96$

Step 5: Conclusion :

At 5% level, the calculated value of $z >$ tabulated value of z.

$\therefore$ Reject H_0. The samples cannot be regarded from the same population.

Example 4.4.17. The average marks scored by 32 boys is 72 with a SD of 8, while that for 36 girls is 70 with a SD of 6. Test at 1% level of significance whether the boys performance better than girls.

Nov.'15, Apr.'15

Solution . Step 1: Given $n_1 = 32$, $n_2 = 36$ and $\overline{x}_1 = 72$, $\overline{x}_2 = 70$ and $s_1 = 8$, $s_2 = 6$.

Step 2 : Null hypothesis (H_0) **:** The performance of boys and girls are equal. i.e., $\mu_1 = \mu_2$

Alternative hypothesis (H_1) **:** $\mu_1 > \mu_2$. $\mu_1 \neq \mu_2$

Step 3: Calculated value of z :

$$\text{Test statistic } z = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{72 - 70}{\sqrt{\frac{(8)^2}{32} + \frac{(6)^2}{36}}}$$

$$\therefore z = 1.1547$$

Step 4: Tabulated value of z at 1% level of significance $= \cancel{2.33}$ 2.58

Step 5: Conclusion :

At 1% level, the calculated value of $z <$ tabulated value of z.

$\therefore$ Accept H_0. Both Performance are equal.

Step 5: Conclusion :

At 5% level, the calculated value of $z >$ tabulated value of z.

$\therefore$ Reject H_0. The mean number of accidents happen in the cities are significant.

Example 4.4.21. A simple sample of heights of 6400 English men has a mean of 170 cm and an SD of 6.4 cm, whole a simple sample of heights of 1600 Americans has a mean of 172 cm and an SD if 6.3 cm. Do the data indicate that American's and on the average, taller than the Englishmen? **Apr.'13, Nov.'13**

Solution . Step 1: Given $n_1 = 6400$, $n_2 = 1600$ and $\overline{x}_1 = 170$, $\overline{x}_2 = 172$ and $s_1 = 6.4$, $s_2 = 6.3$.

Step 2 : Null hypothesis (H_0) : The American's and Englishmen have the same mean height. i.e., $\mu_1 = \mu_2$

Alternative hypothesis (H_1) : $\mu_1 > \mu_2$.

Step 3: Calculated value of z :

$$\text{Test statistic } z = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{170 - 172}{\sqrt{\frac{(6.4)^2}{6400} + \frac{(6.3)^2}{1600}}}$$

$$\therefore z = -11.3216$$

$\therefore$ The calculated value of $|z| = 11.3216$

Step 4: Tabulated value of z at 5% level of significance = ~~1.645~~ 1.96

Step 5: Conclusion :

At 5% level, the calculated value of $z >$ tabulated value of z.

$\therefore$ Reject H_0. Thus The American's and Englishmen have not the same mean height.

4.4.2 Test of Significance of Population and Sample S.D.

Let s be standard deviation of the drawn sample of size n and σ be the standard deviation of the population. Then the test statistic

$$z = \frac{s - \sigma}{\sigma/\sqrt{2n}}.$$

Working rule :

1. Write the sample size n, sample S.D.(s) and population S.D.(σ).
2. State null hypothesis H_0 and alternative hypothesis H_1.
3. Find the calculated value of z by using test statistic $z = \frac{s-\sigma}{\sigma/\sqrt{2n}}$. and consider $|z|$ if z becomes negative.
4. Write tabulated value of z at 1% or 5% level of significance.
5. Conclusion :

   (i) If the calculated value of $z <$ tabulated value of z, then accept H_0 (or)

(ii) If the calculated value of $z >$ tabulated value of z, then reject H_0.

Example 4.4.27. A manufacturer of electric bulbs, according to a certain process, find the standard deviation of the life of lamps to be 100 hours. He wants to change the process, if the new process results in a smaller variation in the life of lamps. In adopting a new process, a sample of 150 bulbs gave a standard deviation of 95 hours. Is the manufacturer justified in changing the processes?

Nov.'06, Apr.'12, Apr.'15

Solution . Step 1: Given sample size $n = 150$, sample S.D. $s = 95$ and population S.D. $\sigma = 100$.

Step 2 : Null hypothesis (H_0) : $\sigma = 100$

Alternative hypothesis (H_1) : $\sigma \neq 100$.

Step 3: Calculated value of z :

$$\text{Test statistic } z = \frac{s - \sigma}{\sigma/\sqrt{2n}} = \frac{95 - 100}{95/\sqrt{2 \times 150}} = -0.866$$

$\therefore$ The calculated value of $|z| = 0.866$

Step 4: Tabulated value of z at 5% level of significance $= 1.96$

Step 5: Conclusion :

At 5% level, the calculated value of $z <$ tabulated value of z.

$\therefore$ Accept H_0. Changing of process is not significant.

Example 4.4.28. A random sample of 60 observations were selected from a large population and its standard deviation was found to be 2.5. Can it be concluded that this sample is taken from a population with S.D. 3?

Solution . Step 1: Given $n = 60$, and sample S.D. $s = 2.5$ and population S.D.$\sigma = 3$.

Step 2 : **Null hypothesis (H_0) :** $\sigma = 3$

Alternative hypothesis (H_1) : $\sigma \neq 3$.

Step 3: Calculated value of z **:**

$$\text{Test statistic } z = \frac{s - \sigma}{\sigma/\sqrt{2n}} = \frac{2.5 - 3}{3/\sqrt{2 \times 60}}$$

$$z = -1.826$$

$\therefore$ The calculated value of $|z| = 1.826$

Step 4: Tabulated value of z at 5% level of significance = 1.96

Step 5: Conclusion :

At 5% level, the calculated value of $z <$ tabulated value of z.

$\therefore$ Accept H_0. $\therefore$ Sample belongs to the population with S.D. 3.

4.4.3 Test of Significance of The Difference of Standard Deviations

Let s_1 and s_2 be the standard deviations of two independent samples n_1 and n_2 drawn from the two populations with standard deviations σ_1 and σ_2 respectively. Then testing of the samples have come from two populations by using the test statistic

$$z = \frac{s_1 - s_2}{\sqrt{\frac{\sigma_1^2}{2n_1} + \frac{\sigma_2^2}{2n_2}}}.$$

If σ_1 and σ_2 are not known then the test statistic

$$z = \frac{s_1 - s_2}{\sqrt{\frac{s_1^2}{2n_1} + \frac{s_2^2}{2n_2}}}.$$

Working rule :

1. Write the samples sizes n_1, n_2 and S.D.s s_1, s_2.
2. State null hypothesis H_0 and alternative hypothesis H_1.

3. Find the calculated value of z by using test statistic $z = \frac{s_1 - s_2}{\sqrt{\frac{s_1^2}{2n_1} + \frac{s_2^2}{2n_2}}}$. and consider $|z|$ if z becomes negative.
4. Write tabulated value of z at 1% or 5% level of significance.
5. Conclusion :

   (i) If the calculated value of $z <$ tabulated value of z, then accept H_0 (or)

   (ii) If the calculated value of $z >$ tabulated value of z, then accept H_1.

Example 4.4.29. The standard deviation of a random sample of 900 members is 4.6 and that of another independent sample of 1600 members is 4.8. Examine if the two samples could have been drawn from a population with standard deviation 4.0?

Nov.'10

Solution . Step 1: Given samples sizes $n_1 = 900$, $n_2 = 1600$ and sample S.D.s $s_1 = 4.6$, $s_2 = 4.8$

Step 2 : Null hypothesis (H_0) **:** $\sigma_1 = \sigma_2$

Alternative hypothesis (H_1) **:** $\sigma_1 \neq \sigma_2$.

Step 3: Calculated value of z **:**

$$\text{Test statistic } z = \frac{s_1 - s_2}{\sqrt{\frac{s_1^2}{2n_1} + \frac{s_2^2}{2n_2}}} = \frac{4.6 - 4.8}{\sqrt{\frac{(4.6)^2}{2\times 900} + \frac{(4.8)^2}{2\times 1600}}}$$

$$\therefore z = -1.453$$

$\therefore$ The calculated value of $|z| = 1.453$

Step 4: Tabulated value of z at 5% level of significance $= 1.96$

Step 5: Conclusion :

At 5% level, the calculated value of $z <$ tabulated value of z.

$\therefore$ Accept H_0. Thus the difference between the sample standard deviations are not significant.

Example 4.4.30. The standard deviation of a population is 3. Sample of 1000 and 500 are drawn from it and their standard deviations are respectively 2.6 and 2. Is the difference between standard deviations significant? **Nov.'14**

Solution . Step 1: Given samples sizes $n_1 = 1000$, $n_2 = 500$ and sample S.D.s $s_1 = 2.6$, $s_2 = 2$

Step 2 : Null hypothesis (H_0) : $\sigma_1 = \sigma_2$
Alternative hypothesis (H_1) : $\sigma_1 \neq \sigma_2$.

Step 3: Calculated value of z **:**

$$\text{Test statistic } z = \frac{s_1 - s_2}{\sqrt{\frac{s_1^2}{2n_1} + \frac{s_2^2}{2n_2}}} = \frac{2.6 - 2}{\sqrt{\frac{(2.6)^2}{2\times1000} + \frac{(2)^2}{2\times500}}}$$

$$\therefore z = 6.984$$

Step 4: Tabulated value of z at 5% level of significance = 1.96

Step 5: Conclusion :
At 5% level, the calculated value of $z >$ tabulated value of z.
$\therefore$ Reject H_0. Thus the difference between the sample standard deviations are significant.

Example 4.4.31. In two samples of sizes 150 and 250 the standard deviations were calculated as 15.3 and 13.8. Can we conclude that both the samples are drawn from the populations with the same standard deviations ?

Solution . Step 1: Given samples sizes $n_1 = 150$, $n_2 = 250$ and sample S.D.s $s_1 = 15.3$, $s_2 = 13.8$

Step 2 : Null hypothesis (H_0) : $\sigma_1 = \sigma_2$
Alternative hypothesis (H_1) : $\sigma_1 \neq \sigma_2$.

Step 3: Calculated value of z :

$$\text{Test statistic } z = \frac{s_1 - s_2}{\sqrt{\frac{s_1^2}{2n_1} + \frac{s_2^2}{2n_2}}} = \frac{15.3 - 13.8}{\sqrt{\frac{(15.3)^2}{2\times150} + \frac{(13.8)^2}{2\times250}}}$$

$$\therefore z = 1.392$$

Step 4: Tabulated value of z at 5% level of significance $= 1.96$

Step 5: Conclusion :

At 5% level, the calculated value of $z <$ tabulated value of z.

$\therefore$ Reject H_0. Thus the difference between the sample standard deviations are not significant.

Example 4.4.32. Two machines A and B produced 200 and 250 items on the average per day with a S.D. of 20 and 25 items respectively on the basis of records of 50 day's production. Can you regard both machines equally efficient at 1% level of significance.

Solution . Step 1: Given sample size $n_1 = 200 \times 50 = 10000$, $n_2 = 250 \times 50 = 12500$ and sample S.D.s $s_1 = 20$, $s_2 = 25$

Step 2 : Null hypothesis (H_0) : $\sigma_1 = \sigma_2$

Alternative hypothesis (H_1) : $\sigma_1 \neq \sigma_2$.

Step 3: Calculated value of z :

$$\text{Test statistic } z = \frac{s_1 - s_2}{\sqrt{\frac{s_1^2}{2n_1} + \frac{s_2^2}{2n_2}}} = \frac{20 - 25}{\sqrt{\frac{(20)^2}{2\times10000} + \frac{(25)^2}{2\times12500}}}$$

$$\therefore z = -23.57$$

$\therefore$ The calculated value of $|z| = 23.57$

Step 4: Tabulated value of z at 1% level of significance $= 2.58$

Step 5: Conclusion :

At 1% level, the calculated value of $z >$ tabulated value of z.

$\therefore$ Reject H_0. Thus the both machines are not equally efficient at 1% level.

Example 4.4.33. From the given data, find out whether the difference between the standard deviations of the two samples is a real one. $n_1 = 1392$, $\sigma_1 = 53.84$, $n_2 = 630$ and $\sigma_2 = 56.56$.

Solution . Step 1: Given sample size $n_1 = 1392$, $\sigma_1 = 53.84$, $n_2 = 630$ and $\sigma_2 = 56.56$.

Step 2 : Null hypothesis (H_0) : $\sigma_1 = \sigma_2$ i.e., The difference is not significant.

Alternative hypothesis (H_1) : $\sigma_1 \neq \sigma_2$.

Step 3: Calculated value of z :

$$\text{Test statistic } z = \frac{s_1 - s_2}{\sqrt{\frac{s_1^2}{2n_1} + \frac{s_2^2}{2n_2}}} = \frac{53.84 - 56.56}{\sqrt{\frac{(53.84)^2}{2\times1392} + \frac{(56.56)^2}{2\times630}}}$$

$$\therefore z = -1.44$$

$\therefore$ The calculated value of $|z| = 1.44$

Step 4: Tabulated value of z at 1% level of significance $= 1.92$

Step 5: Conclusion :

At 1% level, the calculated value of $z <$ tabulated value of z.

$\therefore$ Accept H_0. Thus the differences are not significant.

Example 4.4.34. The standard deviation of a random sample of 900 members is 5 years and of another independent sample of 1000 members is 4.5 years. Can the samples be regarded as drawn from equally variable normal population.

Solution . Step 1: Given sample size $n_1 = 900$, $n_2 = 1000$ and sample S.D.s $s_1 = 5$, $s_2 = 4.5$

Step 2 : Null hypothesis (H_0) : $\sigma_1 = \sigma_2$

Alternative hypothesis (H_1) : $\sigma_1 \neq \sigma_2$.

Step 3: Calculated value of z :

$$\text{Test statistic } z = \frac{s_1 - s_2}{\sqrt{\frac{s_1^2}{2n_1} + \frac{s_2^2}{2n_2}}} = \frac{5 - 4.5}{\sqrt{\frac{(5)^2}{2\times 900} + \frac{(4.5)^2}{2\times 1000}}}$$

$$\therefore z = 3.23$$

Step 4: Tabulated value of z at 1% level of significance $= 2.58$

Step 5: Conclusion :

At 1% level, the calculated value of $z >$ tabulated value of z.

$\therefore$ Reject H_0. Samples are not drawn from same population.

Exercise

1. For two samples, the following data are given

$n_1 = 1000 \quad \bar{x}_1 = 67.42 \quad s_1 = 2.58$

$n_2 = 1200 \quad \bar{x}_2 = 67.25 \quad s_2 = 2.50$ **Nov.'05**

Is the difference between the standard deviations significant?

Ans. $z = 1.039$ Accept H_0. Sample standard deviations do not differ significantly.

2. Intelligence tests of two groups of boys and girls gave the following results.

Boys : $\bar{x}_1 = 86 \quad s_1 = 11 \quad n_1 = 150$

Girls : $\bar{x}_2 = 88 \quad s_2 = 10 \quad n_2 = 125$

Is the difference between the standard deviations significant?

Ans. $z = 1.116$ Accept H_0. The difference between the standard deviations is not significant.

4.5 Part – A : Two Marks Questions

Question 92. Write down the principle of least squares.
Nov.'12, Apr.'13, Apr.'15, Nov.'15

Answer . The best fit of the curve is that for which the residual error at each point of x_i for the given pairs (x_i, y_i) are as small as possible. Then the sum of squares of the errors is a minimum. This is known as the principle of least squares.

Question 93. Define curve fitting.

Answer . Curve fitting is the process of constructing a curve, or mathematical function, that has the best fit to a series of data points, possibly subject to constraints. (or)

Curve fitting is a mathematical model described by equations or systems of equations to observations in order to estimate parameters that can be used to interpret the experimental data.

Question 94. What is the error committed, when fitting a straight line of the form $y = ax + b$ for the observed values $(x_i, y_i) : i = 1, 2, 3, ...n$. **Apr.'12**

Answer . When we fit a straight line $y = ax + b$ for given observed values (x_i, y_i) ; $i = 1, 2, ...n$ then $y - y_i$ is a error called residual error.

Question 95. Write the normal equations to fit a curve the straight line $y = ax + b$ by the method of least squares.

Solution . The normal equations to fit a curve in the straight line $y = ax + b$ is

$$a\Sigma x + nb = \Sigma y$$
$$a\Sigma x^2 + b\Sigma x = \Sigma xy$$

Question 96. Write down the normal equations to fit the parabola $y = ax^2 + bx + c$ for the method of least squares. **Apr.'11, Nov.'15**

Answer . The normal equations of the parabolic fit of the curve are

$$\begin{aligned} a\Sigma x^2 + b\Sigma x + nc &= \Sigma y \\ a\Sigma x^3 + b\Sigma x^2 + c\Sigma x &= \Sigma xy \\ a\Sigma x^4 + b\Sigma x^3 + c\Sigma x^2 &= \Sigma x^2 y \end{aligned}$$

Question 97. Write the normal equations to fit a curve of the form $y = ax^b$ by the method of least squares. **Nov.'11**

Answer . The given fit of the curve is $y = ax^b$. ...(1)

Taking logarithm on both sides of (1), we get

$$\begin{aligned} \log_{10} y &= \log_{10}\left(ax^b\right) \\ &= \log_{10} a + \log_{10} x^b \\ \therefore \log_{10} y &= \log_{10} a + b\log_{10} x \\ \text{i.e., } Y &= A + bX \qquad \text{...(2)} \end{aligned}$$

Then we determine the parameters A and B by using method of least squares. The normal equations are

$$\begin{aligned} nA + b\Sigma X &= \Sigma Y \\ A\Sigma X + b\Sigma X^2 &= \Sigma XY. \end{aligned}$$

Question 98. Define 'Population'. **Nov.'14**

Solution . A population consists of collection of individuals, which may be person or experimental outcomes, whose characteristic are to be studied.

Question 99. Define sample.

Solution . A sample is a portion of the population that is studied to learn about the characteristics of the population.

Question 100. Define 'Large and Small samples. **Nov.13, Nov.'14, Nov.'15**

Solution . If the number of elements (or) items (say n) in the sample is greater than or equal to 30 (i.e., $n \geq 30$), then the sample is called as large samples. If the number of elements (or) items (n) in the sample is less than 30 (i.e.,$n < 30$), then it is called as small samples.

Question 101. Define a residual. **Nov.'13**

Solution . The residual (of an observed value) is the difference between the observed value and the estimated function value.

Question 102. Define standard error.

Solution . The standard deviation of the sampling distribution of a statistic is called standard error. Standard error is used to access the difference between the expected and observed values.

Question 103. What is the formula for finding the difference between the sample proportion and population proportion? **Apr.'13**

Answer . Let P is the population proportion and p is the sample proportion then

$$z = \frac{p-P}{\sqrt{\frac{PQ}{n}}} \text{ and } Q = 1-P$$

Question 104. Define hypothesis.

Solution . A hypothesis is a statement of a relationship between two or more variables. A statistical hypothesis is simply a particular kind of hypothesis.

Question 105. Define Statistical hypothesis.

Solution . A statistical hypothesis is either

1. A statement about the value of a population parameter (e.g., mean, median, mode, variance, standard deviation etc), or

2. A statement about the kind of probability distribution that a certain variable obeys.

Question 106. Define "test statistic."

Solution . The test statistic is a mathematical formula to determine the likelihood of obtaining sample outcomes if the null hypothesis were true. The value of the test statistic is used to make a decision regarding the null hypothesis.

Question 107. What are the null and alternative hypothesis?
Nov.'10, Nov.'11, Apr.'12, Apr.'14

Solution . Null hypothesis always predicts that there is no relationship between the variables being studied and it is denoted by H_0. The null hypothesis typically corresponds to a general or a default position. Making this assertion will make no difference and hence cannot be proven positively.

An alternative hypothesis (H_1) is a statement that directly contradicts a null hypothesis by stating that that the actual value of a population parameter is less than, greater than, or not equal to the value stated in the null hypothesis.

Question 108. Define parameters and statistics.

Solution . Parameters are the measures such as mean, standard deviation etc which describe a population and the word statistic is used is used to indicate various measures relating to the sample such as mean, S.D., etc. A statistic is any function of observations in a random sample.

Question 109. Define sampling distribution

Solution . The probability distribution of a statistic is called as a sampling distribution

Question 110. Define critical region or region of rejection.

Solution . A region corresponding to a statistic, in the sample space which amounts to rejection of null hypothesis H_0 is called as critical region. The region of the sample space which amounts to the acceptance of null hypothesis is called acceptance region.

Question 111. Define critical value (or) significant value.

Solution . The value of the test statistic which separates the acceptance region and critical region is called the critical value.

Question 112. What is one tailed test? **Apr.'14**

Solution . A one-tailed test uses an alternate hypothesis that states either $H_1 : \mu > \mu_0$ or $H_1 : \mu < \mu_0$, but not both. One tailed test is also classified as (i) right tailed test (ii) left tailed test.

Question 113. Define right-tail test and left-tail test.

Solution . If the observed test values fall to the right of the critical value then the test is called as right-tail test. In a right-tail test, we accept the null hypothesis if the observed test value is less than or equal to the critical value.

It the observed test values fall to the left of the critical value then the test is called as left-tail test. In a left-tail test, we accept the null hypothesis if the observed test value is greater than or equal to the critical value.

Question 114. Define two tailed test.

Solution . When two tails of the sampling distribution of the normal curve are used, then the test is called as two-tailed test. In the two tailed test the alternative hypothesis is stated as $\neq$.

Question 115. Define Test of significance. **Apr.'15**

Solution . Significance testing (also called hypothesis testing) is a mathematical method which is used to decide whether or not an hypothesis (an assumption made about a population parameter) should be accepted or rejected based upon our sample statistic.

Question 116. Define 'Level of Significance'. **Apr.'11, Nov.'15**

Solution . The level of significance is the maximum probability of committing a type I error. This probability is symbolized by α. That is, P(type I error) $= \alpha$. and P(type II error) $= \beta$.

It is the probability level below which the null hypothesis is rejected. Generally 5% and 1% level of significance are used.

Question 117. What are the Type I and Type II errors in sampling theory? **Nov.'11, Apr.'11, Nov.'15**

Solution . Type I error is the probability of rejecting a null hypothesis (H_0) when it is actually true. Type II error is the probability of accepting a null hypothesis (H_0) when it is false.

Question 118. A sample of 900 items has mean 3.4 and standard deviation 2.61. Can the sample regarded as drawn from a population with mean 3.25 at 5% level of significance? **Apr.'11**

Solution . Given $n = 900$, $\overline{x} = 3.4$, $\mu = 3.25$ and $\sigma = 2.61$
Null hypothesis (H_0) : $\mu = 3.25$.
Alternative hypothesis (H_1) : $\mu \neq 3.25$.

$$\therefore \quad z = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{3.4 - 3.25}{2.61/\sqrt{900}} = 1.724$$

Tabulated value of z at 5% level of significance $= 1.96$
At 5% level, the calculated value of $z <$ tabulated value of z.
$\therefore$ Accept H_0. Difference is not significant.

Question 119. A sample of 900 members from a normal population with SD 2.61 cms has a mean 3.5 cms. Find the 95% fiducial limits for the population mean. **Nov.'10**

Solution . Given $n = 900$, $\overline{x} = 3.5$ and $\sigma = 2.61$

If the population mean μ is unknown, 95% confidence limits of μ are

$$= \overline{x} \pm 1.96 \frac{\sigma}{\sqrt{n}} = 3.5 \pm 1.96 \times \frac{2.61}{\sqrt{900}}$$

$$= 3.5 \pm 0.17 = (3.33,\ 3.67)$$

Question 120. Write down the test statistic for large sample test for difference of means. **Nov.'11**

Solution . If the samples are drawn from the same population with known S.D σ then the test statistic is

$$z = \frac{\overline{x_1} - \overline{x_2}}{\sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Question 121. A coin was tossed 400 times and the head turned up 216 times. Test the hypothesis that the coin is unbiased **Nov.'12**

Solution . Given that, sample size $n = 400$. Number of heads $= 216$.

$p =$ sample proportion of heads $= \frac{216}{400} = 0.54$

$P =$ Population proportion $= \frac{1}{2} = 0.5$. $\therefore$ $Q = 1 - P = 0.5$

Null hypothesis (H_0) : Coin is unbiased. or $P = \frac{1}{2}$

Alternative hypothesis (H_1) : $P \neq \frac{1}{2}$.

$$\therefore z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.54 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{400}}} = 1.6$$

$\therefore$ The calculated value of $z = 1.6$

Tabulated value of z at 5% level of significance $= 1.96$

Since calculated value of $z <$ tabulated value of z.

$\therefore$ Accept H_0. Thus coin is unbiased.